



PH 101: Physics-I

Course Code	PH 101
Course Name	Physics-I
L-T-P-C	3-1-2-5
Prerequisites	NIL

Sem. Both

Equivalent Course:

Course Contents

- **Vector Fields:** Vector transformation, Vector calculus, Divergence and curl in curvilinear coordinates. Divergence and curl of electrostatic fields, Electric potential, Laplace's and Poisson's equation, Divergence and curl of magnetic field, Magnetic vector potential;
- **Electromagnetic Waves:** Maxwell's equations. Conservation of charge and energy, Electromagnetic waves in vacuum and in matter, Polarized and unpolarized electromagnetic waves. Absorption and dispersion;
- **Special Relativity:** Postulates of special relativity, Lorentz transformation, Introduction to four-vectors, Time dilation, Doppler effect, Length contraction, Twin paradox, Relativistic momentum, Mass and energy, energy and momentum, Relativity as bridge between electricity and magnetism, Magnetism as relative phenomenon (Qualitative discussion);
- **Wave Particle Duality and Quantization:** Blackbody radiation, photoelectric effect, Compton effect, DeBroglie waves, Electron Diffraction, Davisson-Germer experiment, Uncertainty principle. Bohr atom model and Sommerfeld corrections. Franck-Hertz experiment.

Practicals

Measurement of magnetic susceptibility by Quinck's method; Determination of Planck's constant by photoelectric effect; Franck-Hertz Experiment; Single-slit, double slits and multiple slits diffraction by Laser; Determination of Planck's constant by radiation method; Stefan's constant; Magnetic field of paired coils in Helmholtz arrangement; Davisson – Germer Experiment

References

1. David J. G., **Introduction to Electrodynamics**, 3rd Edition, Prentice Hall of India
2. Arthur B., **Concepts of Modern Physics**, 6th Edition, Tata McGraw- Hill.
3. Mathew N. O. S., **Elements of Electromagnetic**, 3rd Edition, Oxford University Press
4. Feynman R. P., Leighton R.B. and Sands M., **The Feynman Lectures on Physics**, Volume 1 & II, Narosa Publishing House.